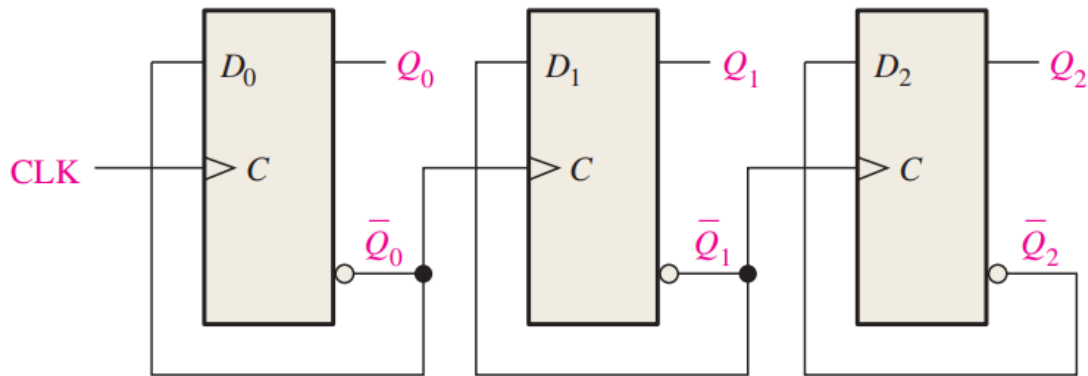


Chapter 9

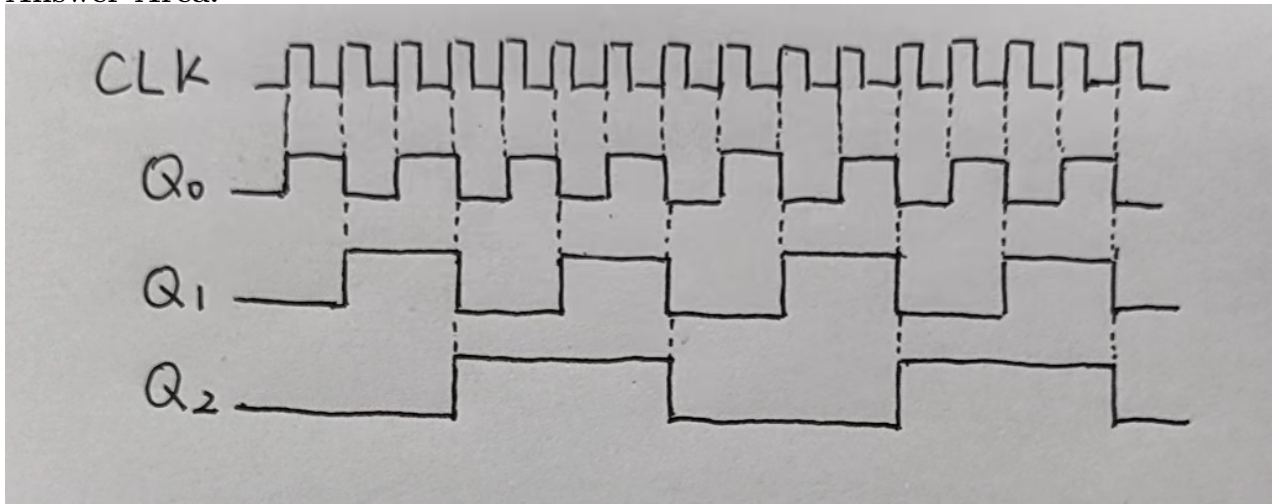
2024.2.22

Name: Chen Zhuoyan Student ID: 523030910037

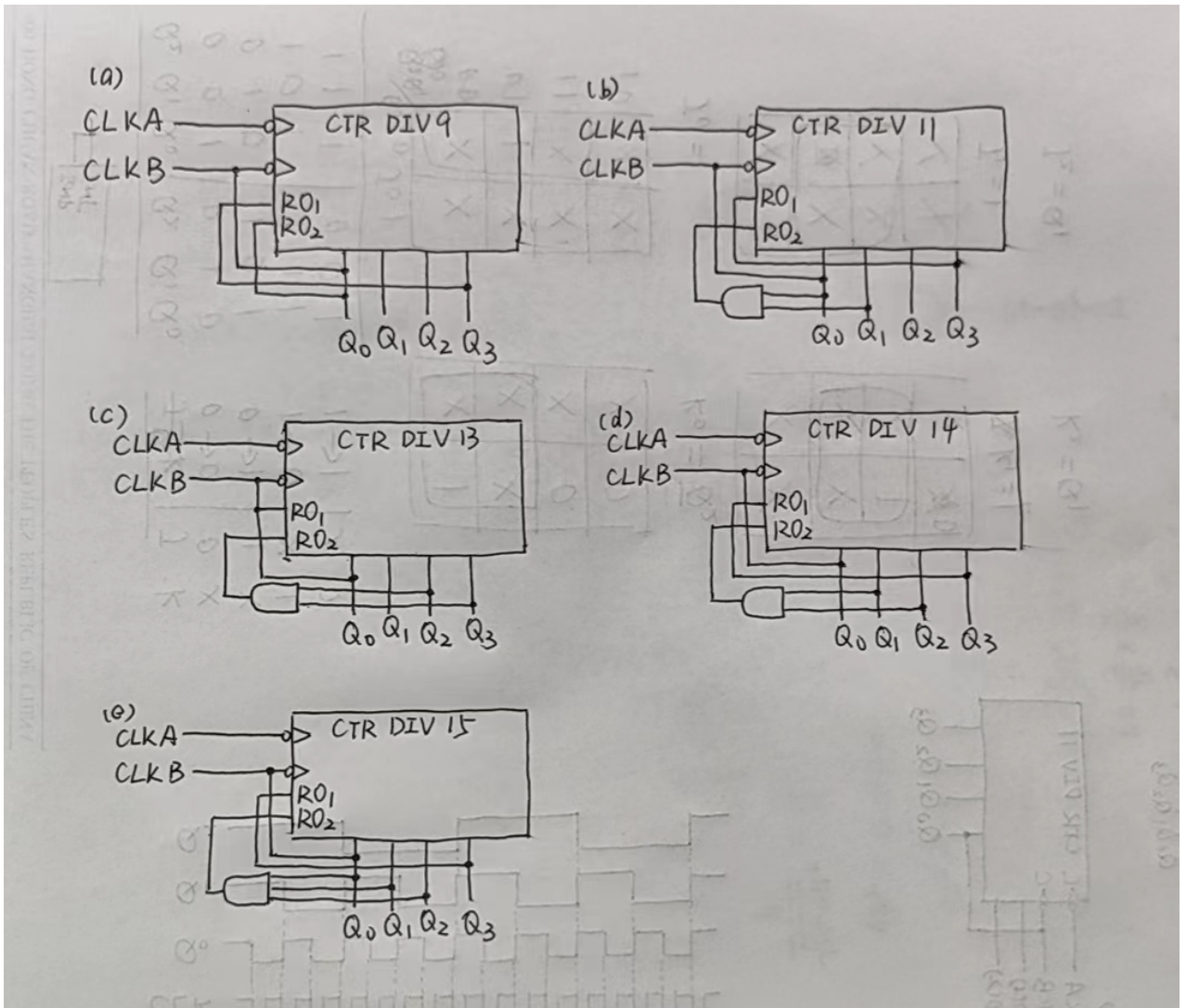
4. For the ripple counter in Figure 9-66, show the complete timing diagram for sixteen clock pulses. Show the clock, Q_0 , Q_1 , and Q_2 waveforms.



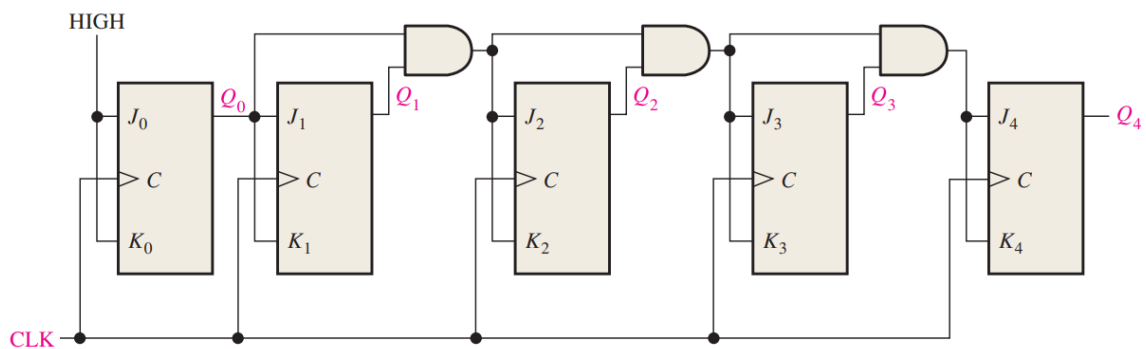
Answer Area:



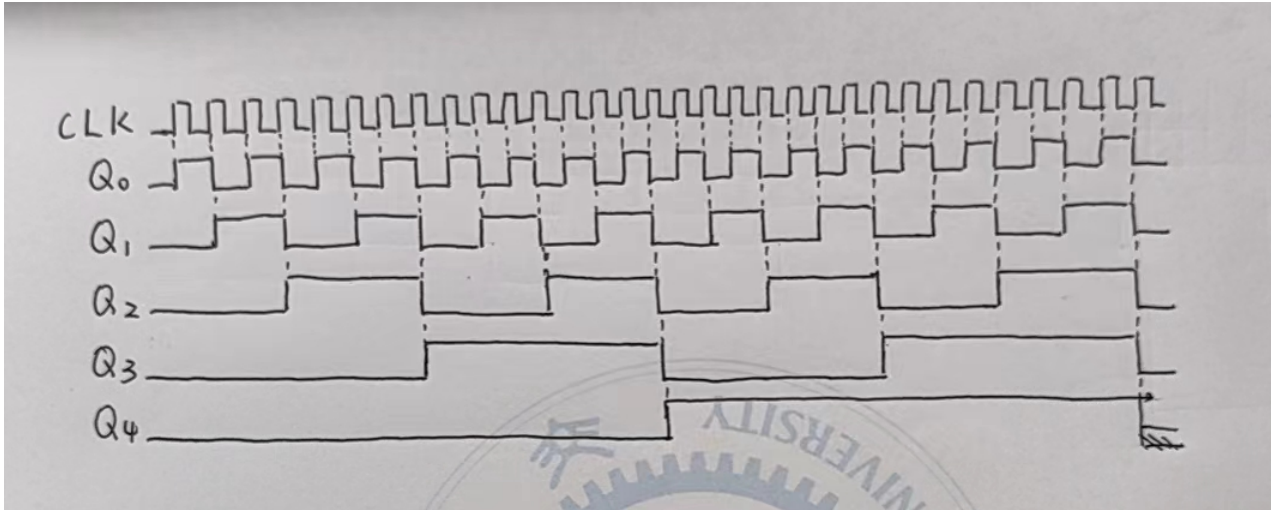
6. Show how to connect a 74HC93 4-bit asynchronous counter for each of the following moduli: (a) 9 (b) 11 (c) 13 (d) 14 (e) 15 **Answer Area:**



8. Show the complete timing diagram for the 5-stage synchronous binary counter in Figure 9-67. Verify that the waveforms of the Q outputs represent the proper binary number after each clock pulse.



Answer Area:



9. By analyzing the J and K inputs to each flip-flop prior to each clock pulse, prove that the decade counter in Figure 9-68 progresses through a BCD sequence. Explain how these conditions in each case cause the counter to go to the next proper state.

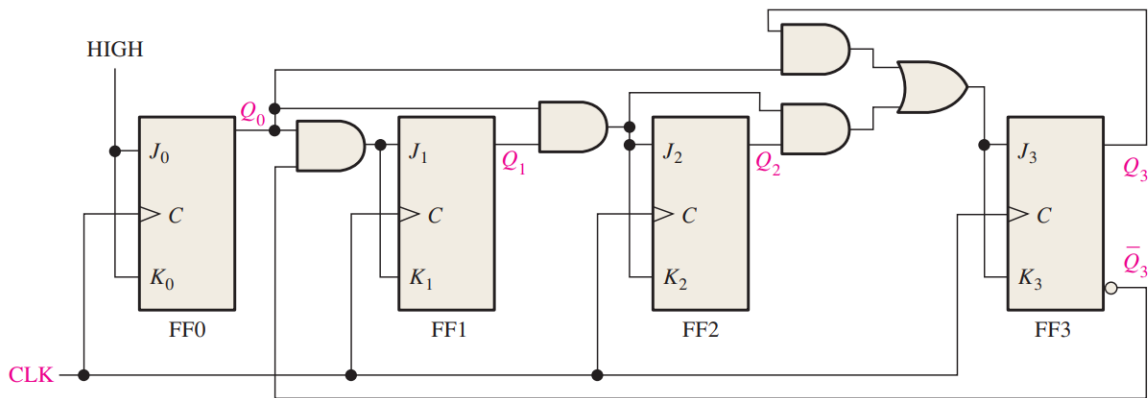


FIGURE 9-68

Answer Area:

$$\because J_0 = K_0 = 1$$

$$J_1 = K_1 = Q_0 \bar{Q}_3$$

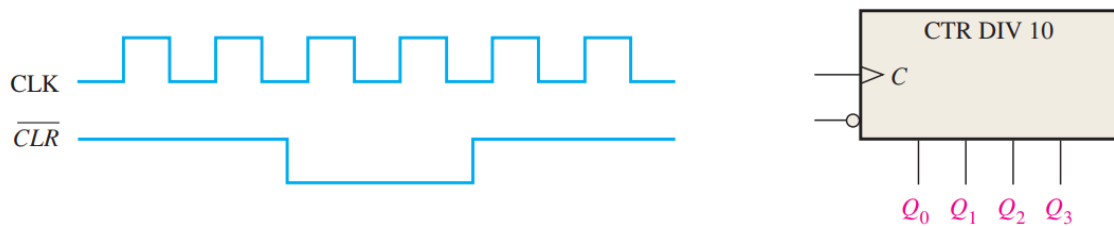
$$J_2 = K_2 = Q_0 Q_1$$

$$J_3 = K_3 = Q_0 Q_1 Q_2 + Q_0 Q_3$$

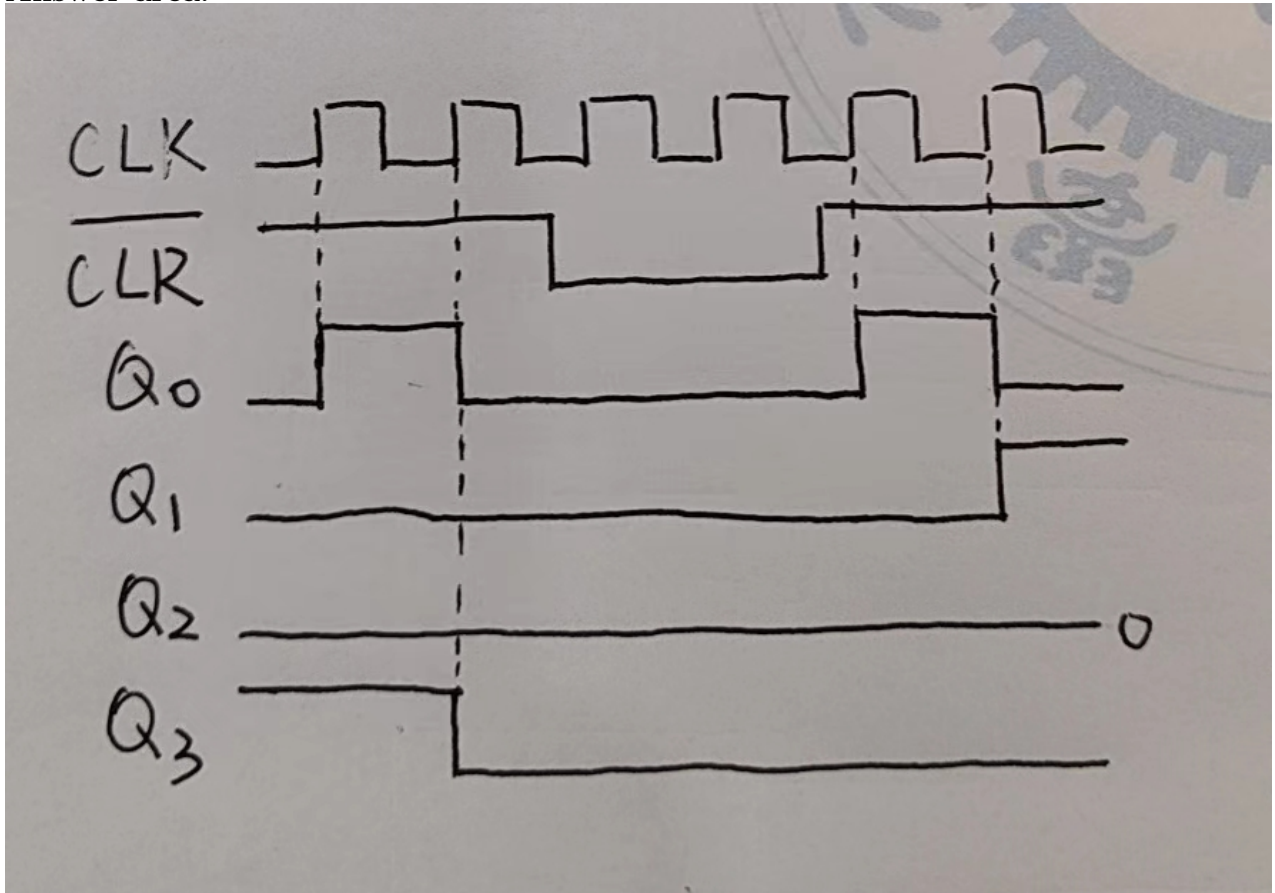
$\therefore$ Assume the flip-flops are initially reset.

CLK	$J_0 K_0$	$J_1 K_1$	$J_2 K_2$	$J_3 K_3$	Q_0	Q_1	Q_2	Q_3
Initially	1	0	0	0	0	0	0	0
1	1	0	0	0	1	0	0	0
2	1	1	0	0	0	1	0	0
3	1	0	0	0	1	1	0	0
4	1	1	1	0	0	0	1	0
5	1	0	0	0	1	0	1	0
6	1	1	0	0	0	1	1	0
7	1	0	0	0	1	1	1	0
8	1	1	1	1	0	0	0	1
9	1	0	0	0	1	0	0	1
10	1	0	0	1	0	0	0	0

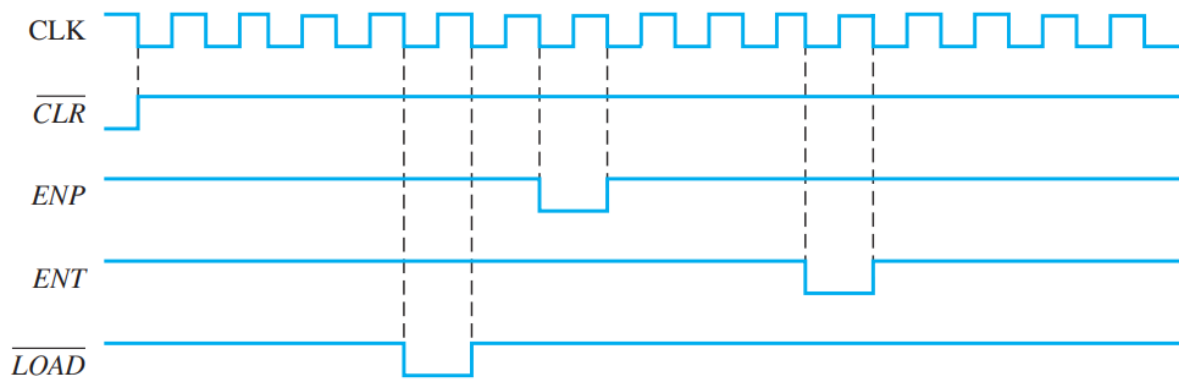
11. A BCD decade counter is shown in Figure 9-70. The waveforms are applied to the clock and clear inputs as indicated. Determine the waveforms for each of the counter outputs (Q_0 , Q_1 , Q_2 , and Q_3). The clear is synchronous, and the counter is initially in the binary 1000 state.



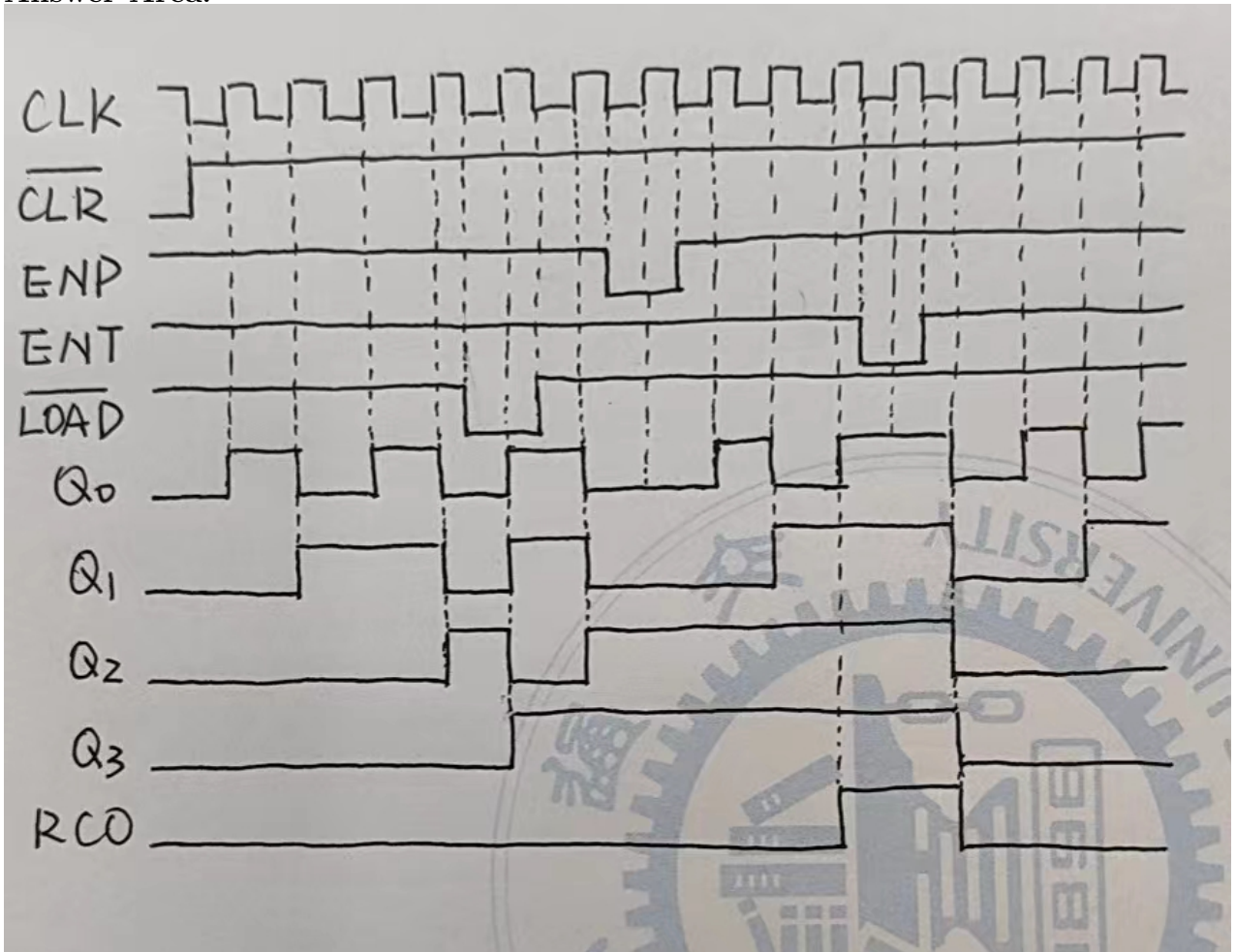
Answer area:



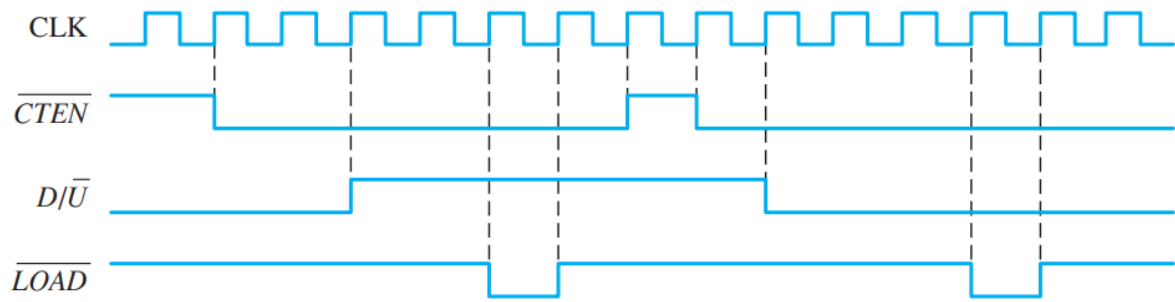
12. The waveforms in Figure 9-71 are applied to a 74HC163 binary counter. Determine the Q outputs and the RCO. The inputs are $D_0 = 1$, $D_1 = 1$, $D_2 = 0$, and $D_3 = 1$.



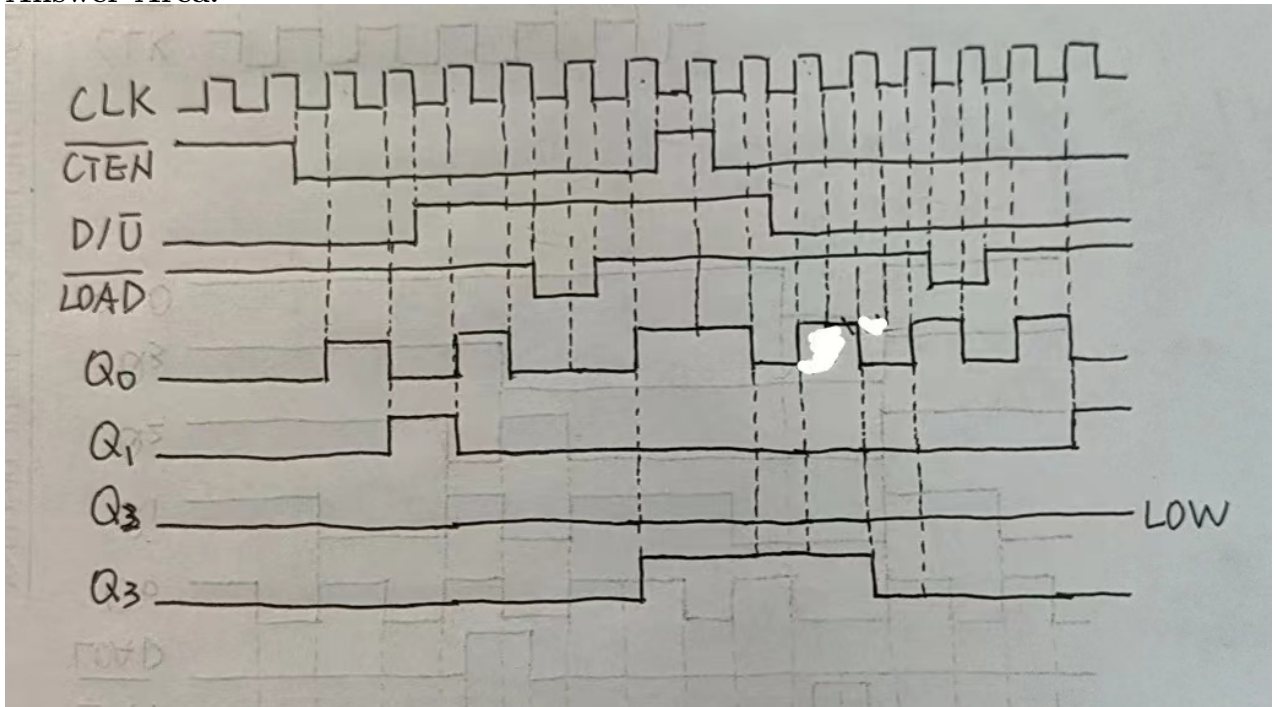
Answer Area:



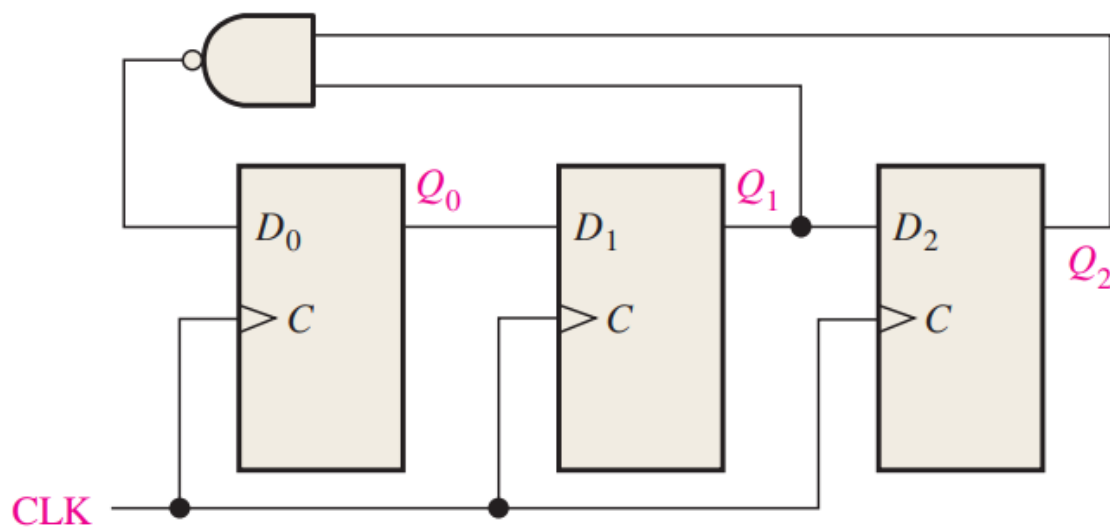
15. Develop the Q output waveforms for a 74HC190 up/down counter with the input waveforms shown in Figure 9-72. A binary 0 is on the data inputs. Start with a count of 0000.



Answer Area:



18. Determine the sequence of the counter in Figure 9-73.



Answer Area:

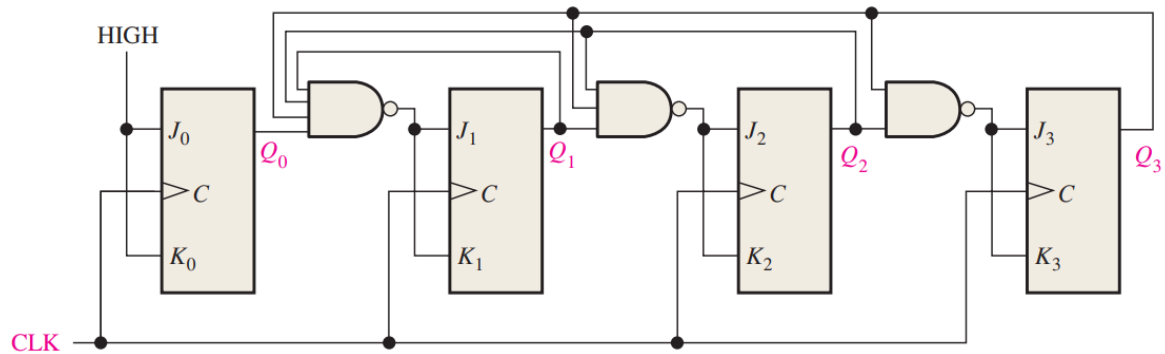
$$\therefore D_0 = \overline{Q_1}Q_2, D_1 = Q_0, D_2 = Q_1$$

$$\therefore$$

CLK	Q_2	Q_1	Q_0	D_2	D_1	D_0
Initially	0	0	0	0	0	1
1	0	0	1	0	1	1
2	0	1	1	1	1	1
3	1	1	1	1	1	0
4	1	1	0	1	0	0
5	1	0	0	0	0	1
6(recycle)	0	0	1	0	1	1

$\therefore$ The sequence is $001 \rightarrow 011 \rightarrow 111 \rightarrow 110 \rightarrow 100 \rightarrow 001 \rightarrow \dots$

19. Determine the sequence of the counter in Figure 9-74. Begin with the counter cleared.



Answer Area:

$$\therefore J_0 = K_0 = 1$$

$$J_1 = K_1 = \overline{Q_0 Q_1 Q_2 Q_3}$$

$$J_2 = K_2 = \overline{Q_1 Q_2 Q_3}$$

$$J_3 = K_3 = \overline{Q_2 Q_3}$$

$$\therefore$$

CLK	Q_0	Q_1	Q_2	Q_3	$J_0 K_0$	$J_1 K_1$	$J_2 K_2$	$J_3 K_3$
Initially	0	0	0	0	1	1	1	1
1	1	1	1	1	1	0	0	0
2	0	1	1	1	1	1	0	0
3	1	0	1	1	1	1	1	0
4	0	1	0	1	1	1	1	1
5	1	0	1	0	1	1	1	1
6(recycle)	0	1	0	1	1	1	1	1

The sequence is $0101 \rightarrow 1010 \rightarrow 0101 \rightarrow \dots$

22. Design a counter to produce the following binary sequence. Use J-K flip-flops.

0, 9, 1, 8, 2, 7, 3, 6, 4, 5, 0, ...

Answer Area:

The next-state table is:

Present State				Next State			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0
0	0	0	0	1	0	0	1
1	0	0	1	0	0	0	1
0	0	0	1	1	0	0	0
1	0	0	0	0	0	1	0
0	0	1	0	0	1	1	1
0	1	1	1	0	0	1	1
0	0	1	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	0	0	0

The transition table for a J-K flip-flop is:

Q_N	Q_{N+1}	J	K
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0

The Karnaugh maps are:

J_0 map

$\therefore J_0 = \bar{Q}_1 \bar{Q}_3 + \bar{Q}_2 \bar{Q}_3$

K_0 map

$K_0 = \bar{Q}_1 \bar{Q}_3 + \bar{Q}_2 \bar{Q}_3$

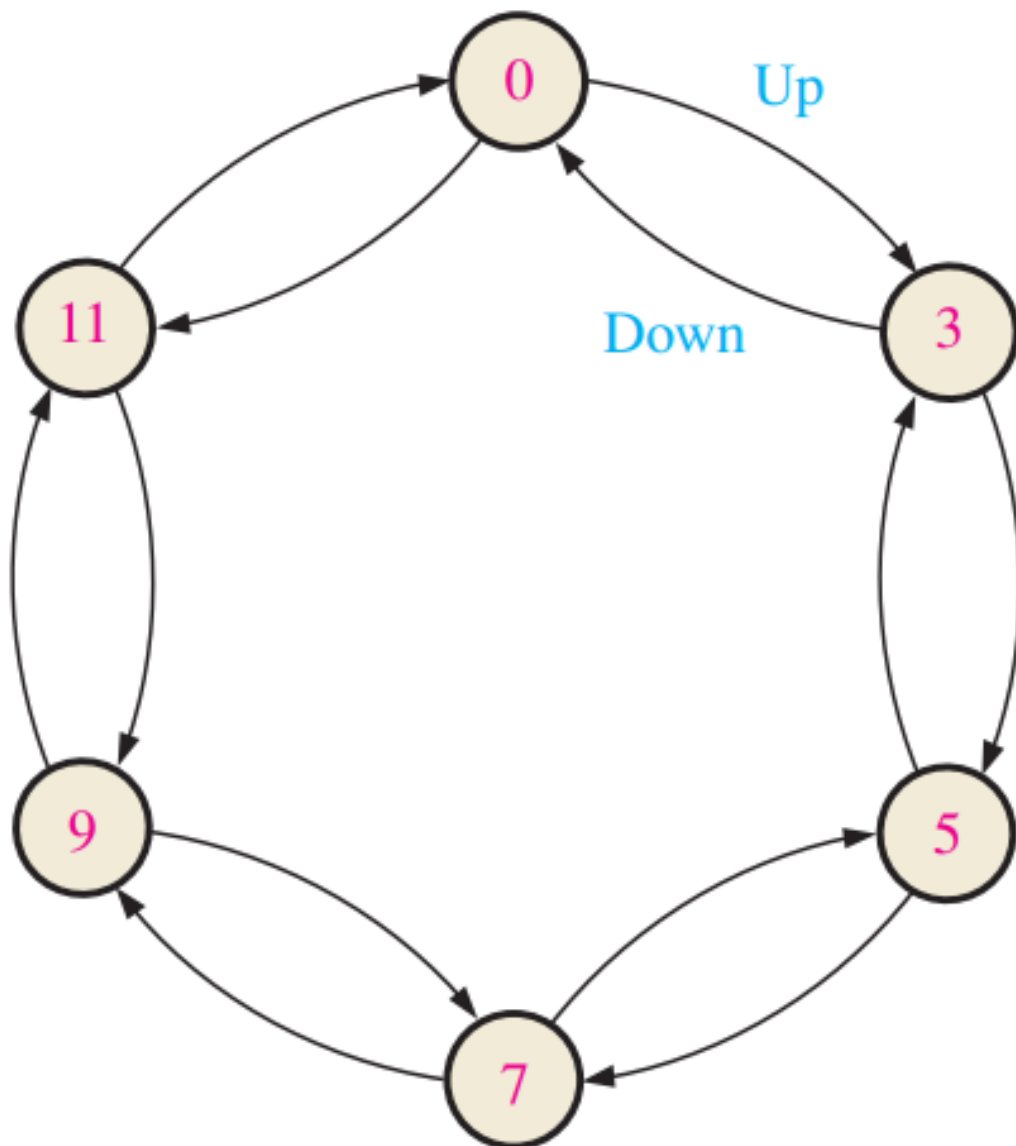
$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	0	0	X	X
01	0	0	X	X
11	X	X	X	X
10	1	0	X	X

$\bar{J}_1$ map
 $\bar{J}_1 = Q_3 \bar{Q}_0$

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	X	X	0	0
01	X	X	0	1
11	X	X	X	X
10	X	X	X	X

K_1 map
 $K_1 = Q_2 \bar{Q}_0$

23.Design a binary counter with the sequence shown in the state diagram of Figure 9–75.



Answer Area:

Suppose $Y=0$ represents count up and $Y=1$ represents count down.

When count up, the next-state table is :

Present State				Next State			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0	0	1	1
0	0	1	1	0	1	0	1
0	1	0	1	0	1	1	1
0	1	1	1	1	0	0	1
1	0	0	1	1	0	1	1
1	0	1	1	0	0	0	0

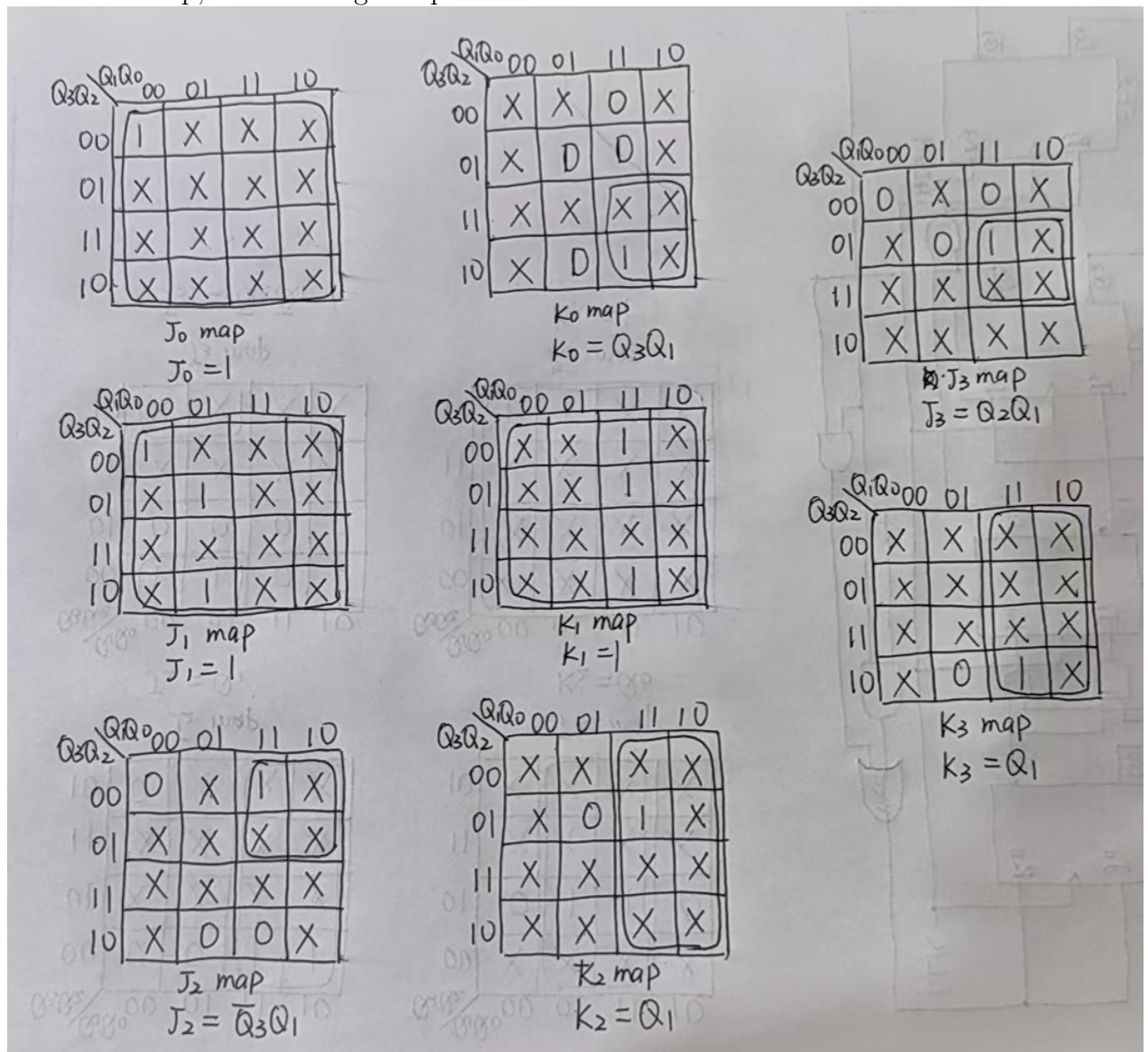
When count down, the next-state table is :

Present State				Next State			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0
0	0	0	0	1	0	1	1
1	0	1	1	1	0	0	1
1	0	0	1	0	1	1	1
0	1	1	1	0	1	0	1
0	1	0	1	0	0	1	1
0	0	1	1	0	0	0	0

The transition table for a J-K flip-flop is:

Q_N	Q_{N+1}	J	K
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0

When count up, the Karnaugh maps are:



When count-down, the Karnaugh maps are:

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	1	X	X	X
01	X	X	X	X
11	X	X	X	X
10	X	X	X	X

J_0 map

$$J_0 = 1$$

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	X	X	1	X
01	X	0	0	X
11	X	X	X	X
10	X	0	0	X

K_0 map

$$K_0 = \bar{Q}_3 \bar{Q}_2$$

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	1	X	X	X
01	X	1	X	X
11	X	X	X	X
10	X	1	X	X

J_1 map

$$J_1 = 1$$

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	X	X	1	X
01	X	X	1	X
11	X	X	X	X
10	X	X	1	X

K_1 map

$$K_1 = 1$$

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	0	X	0	X
01	X	X	X	X
11	X	X	X	X
10	X	1	0	X

J_2 map

$$J_2 = Q_3 \bar{Q}_1$$

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	X	X	X	X
01	X	1	0	X
11	X	X	X	X
10	X	X	X	X

K_2 map

$$K_2 = \bar{Q}_1$$

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	1	X	0	X
01	X	0	0	X
11	X	X	X	X
10	X	X	X	X

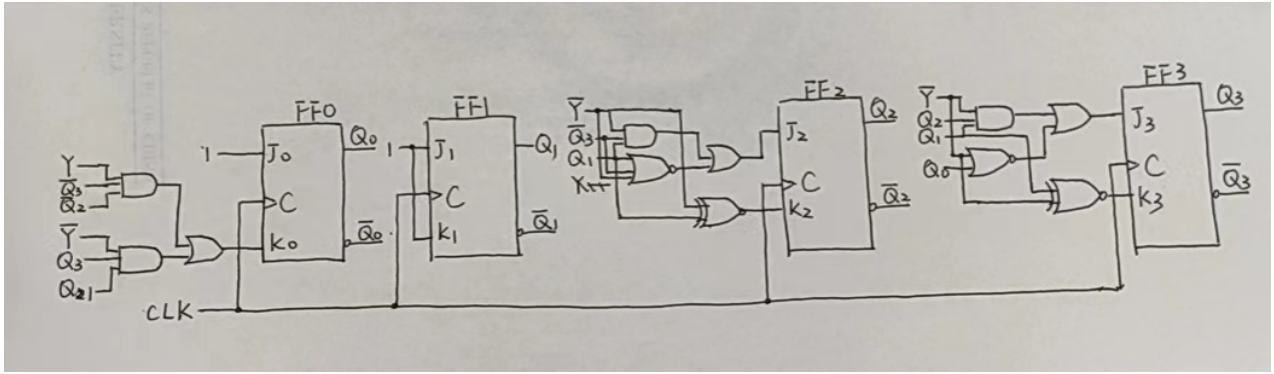
J_3 map

$$J_3 = \bar{Q}_0$$

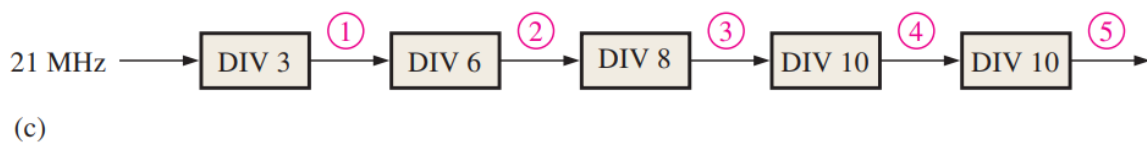
$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	X	X	X	X
10	X	1	0	X

K_3 map

$$K_3 = \bar{Q}_1$$



24. For each of the cascaded counter configurations in Figure 9-76, determine the frequency of the waveform at each point indicated by a circled number, and determine the overall modulus.



Answer Area:

(1) $f = 21/3 = 7 \text{ MHz}$

(2) $f = 7/6 = \frac{7}{6} \text{ MHz} = 1.167 \text{ MHz}$

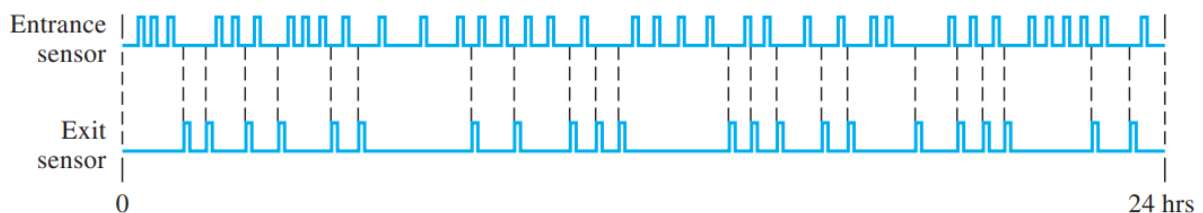
(3) $f = \frac{7}{48} \text{ MHz} = 145.875 \text{ kHz}$

(4) $f = \frac{175}{12} \text{ kHz} = 14.588 \text{ kHz}$

(5) $f = \frac{875}{6} \text{ kHz} = 1.459 \text{ kHz}$

The overall modulus is $3 \times 6 \times 8 \times 10 \times 10 = 14400$

35. For the automobile parking control system in Figure 9-51, a pattern of entrance and exit sensor pulses during a given 24-hour period are shown in Figure 9-78. If there were 53 cars already in the garage at the beginning of the period, what is the state of the counter at the end of the 24 hours?

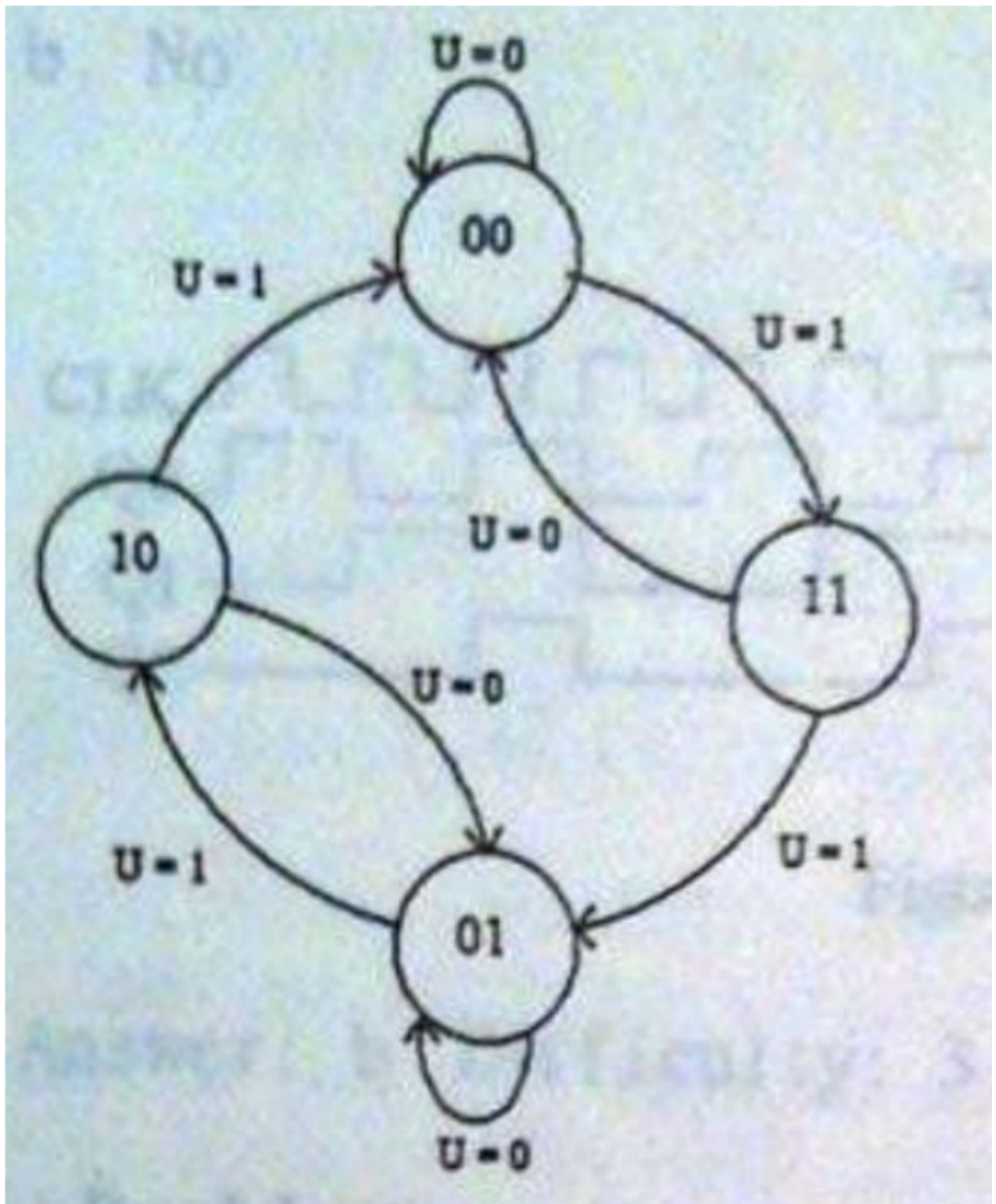


Answer Area:

According to the graph, in the past 24 hours, 37 cars arrived at the parking lot and 22 cars left the parking lot. Therefore at the end of the 24 hours, there are $53 - 22 + 37 = 68$ cars in the parking lot. So the state of the counter is 68.

External problem:

Design a circuit using J-K flip-flops that will perform the count sequence of the state diagram shown below:



Answer Area:

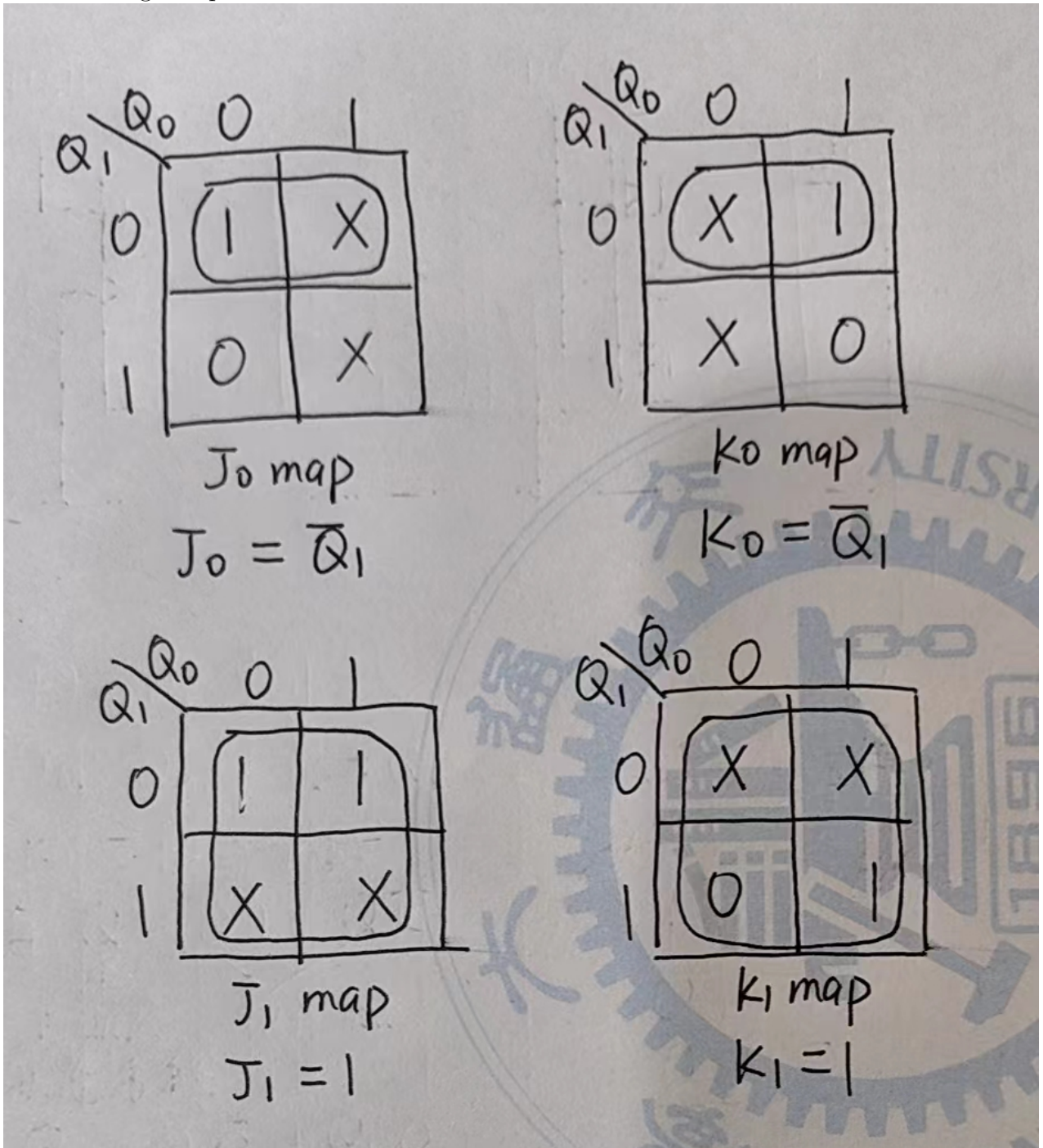
The transition table for a J-K flip-flop is:

Q_N	Q_{N+1}	J	K
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0

When $U = 1$, the next-state table is :

Present State	Next State	Present State	Next State
Q_1	Q_0	Q_1	Q_0
0	0	1	1
1	1	0	0
0	1	0	1
1	0	0	1

The Karnaugh maps are:



When $U = 0$, the next-state table is :

Present State	Next State	Present State	Next State
Q_1	Q_0	Q_1	Q_0
0	0	0	0
1	1	0	1
0	1	1	0
1	0	0	0

The Karnaugh maps are:

$Q_1 \backslash Q_0$	0	1
0	0	X
1	0	X

J_0 map

$$J_0 = 0$$

$Q_1 \backslash Q_0$	0	1
0	X	1
1	X	0

K_0 map

$$K_0 = \bar{Q}_1$$

$Q_1 \backslash Q_0$	0	1
0	0	1
1	X	X

J_1 map

$$J_1 = Q_0$$

$Q_1 \backslash Q_0$	0	1
0	X	X
1	1	1

K_1 map

$$K_1 = 1$$

